Cerebellar Control of Balance and Locomotion

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The cerebellum is important for movement control and plays a particularly crucial role in balance and locomotion. As such, one of the most characteristic signs of cerebellar damage is walking ataxia. It is not known how the cerebellum normally contributes to walking, although recent work suggests that it plays a role in the generation of appropriate patterns of limb movements, dynamic regulation of balance, and adaptation of posture and locomotion through practice. The purpose of this review is to examine mechanisms of cerebellar control of balance and locomotion, emphasizing studies of humans and other animals. Implications for rehabilitation are also considered. NEUROSCIENTIST 10(3):247–259, 2004. DOI: 10.1177/1073858404263517

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In one of the earliest descriptions (2nd century AD), Galen speculated that the cerebellum was the source of the spinal cord and motor nerves and that the cerebellar vermis was a valve to control the flow of animal spirits from the ventricles (Clarke and O'Malley 1996). Views of cerebellar function changed through later centuries and have included control of memory and intellect, involuntary actions such as heartbeat or breathing (Willis, 17th century), and even sexuality (Gall, 19th century; Clarke and O'Malley 1996). In the 19th and 20th centuries, animal lesion experiments and careful observations of humans with cerebellar damage proved that the cerebellum has a significant role in movement control, although the nature of that role was (and still is) debated. Opinions about cerebellar function ranged from a tonic facilitating influence (Rolando, 19th century; Luciani, late 19th century; Holmes, 20th century), to a coordinator of movements produced by other motor structures (Flourens, 19th century; Babinski, early 20th century), to the head proprioceptive ganglion regulating postural tone of muscle groups (Sherrington, early 20th century; Clarke and O'Malley 1996). In this period, scientists also thought that the cerebellum had a unitary function, with no localization of control for different body parts or types of movement.

The modern view is that different cerebellar regions play an integral role in the control of different behaviors including voluntary limb movements, eye movements, balance, locomotion, and now even higher brain (cognitive) functions. This view is based on the anatomy of cerebellar afferent and efferent connections as well as neural recording and lesion studies. It is also clear that the cerebellum exerts control over the flexibility of these behaviors; cerebellar integrity is critical for trial-and-error adaptation of motor behaviors to new contexts. This review will focus on cerebellar contributions to the control and adaptation of balance and locomotion. For information about other important cerebellar topics, we recommend several reviews: cerebellar development and anatomy (Goldowitz and Hamre 1998; Voogd and Glickstein 1998), cerebellar projections (Middleton and Strick 1997; Voogd and Glickstein 1998; Dum and others 2002), plasticity in the cerebellum (Ito 1989, 2001; Hansel and others 2001), learning (Raymond and others 1996; Thach 1997; Ito 2000), and limb movement coordination (Thach and others 1992; Thach 1998).

Clinical Features

Some of the most distinctive clinical signs of cerebellar damage are impairments of balance and gait. Balance abnormalities are characterized by increased postural sway, either excessive or diminished responses to perturbations, poor control of equilibrium during motions of other body parts, and abnormal oscillations of the trunk (titubation). Gait ataxia, or walking incoordination, is often described as a "drunken gait," with distinctive features including variable foot placement, irregular foot trajectories, a widened base of support, a veering path of movement, and abnormal interjoint coordination patterns (Holmes 1917, 1939; Hallett and Massaquoi 1993; Palliyath and others 1998; Crowdy and others 2000; Earhart and Bastian 2001).

Cerebellar Anatomy

To appreciate how the cerebellum might control balance and locomotion, we briefly highlight some aspects of cerebellar anatomy. Most afferent information enters the cerebellum via mossy or climbing fibers; these two fiber types affect cerebellar cortical Purkinje cells differently

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and are thought to transmit distinct types of information. Mossy fibers carry afferent information from the spinal cord, brainstem, and cerebral cortex (via the pons) and set the moment-to-moment firing rate of Purkinje cells (via granule cell parallel fibers; Maekawa and Kimura 1980; Stone and Lisberger 1990). Climbing fibers come from the inferior olive only and cause Purkinje cells to fire only a few complex action potentials (complex spikes) per second (Thach 1968). The firing rate of climbing fibers is thought to be too slow to convey ongoing information about specific movement parameters (Thach 1968; Brooks and Thach 1981; Armstrong and others 1988). Instead, they might be important for motor learning, serving as a teaching (Marr 1969) or error (Albus 1971) signal that adjusts Purkinje cell responsiveness to parallel fiber inputs via long-term depression (Ito 1989; Hansel and others 2001; Ito 2001).

The arrangement of cells in the cerebellar cortex is highly invariant across the entire structure, making it impossible to subdivide the cerebellum based only on cortical anatomy. Instead, the cerebellum is divided into distinct functional zones based on afferent and efferent connectivity (Jansen and Brodal 1940; Ito 1984; Voogd and Glickstein 1998). All zones can influence locomotion, though in different ways. The most medial zone of the cerebellum (vermis) receives input from the primary vestibular afferents and vestibular nuclei (Kotchabhakdi and Walberg 1978), reticular (Clendenin and others 1974a, 1974b) and pontine nuclei, as well as from the spinal cord (Grant 1962; Matshushita and Okado 1981) via dorsal (DSCT) and ventral spinocerebellar tracts (VSCT). It is important to note that the DSCT conveys information about the sensory state of the limbs (Bosco and Poppele 2001), whereas the VSCT also carries information about motor commands to spinal interneurons (Lundberg 1971; Lundberg and Weight 1971; Bosco and Poppele 2001). The medial zone projects out (via the fastigial nuclei) primarily to vestibular and reticular nuclei, although there are also much less dense projections directly to the spinal cord and to the thalamus (Asanuma and others 1983a, 1983c). Thus, the medial cerebellar zone can integrate spinal and vestibular inputs and influence important motor pathways for walking (i.e., vestibulospinal and reticulospinal tracts).

The cerebellar hemispheres can be divided into intermediate and lateral zones. The intermediate zone of the cerebellar hemisphere receives input from the DSCT and VSCT (Grant 1962; Matshushita and Okado 1981), reticular nuclei (Clendenin and others 1974a, 1974b), and cerebral cortical areas (via pontine nuclei; Brodal and Bjaalie 1992, 1997). The intermediate zone projects out (via interpositus in monkey, or globose and emboliform in man) to the red nucleus and to the cerebral cortex via the thalamus (Asanuma and others 1983a, 1983b). Thus, the intermediate zone can integrate spinal and cortical inputs and influence walking through projections to motor cortical areas.

The lateral zone of the cerebellar hemisphere receives input primarily from cerebral cortical areas (via pontine nuclei), with the densest projections from the primary motor, premotor, primary somatosensory, and posterior parietal cortices and also some from prefrontal and temporal lobe areas (Brodal and Bjaalie 1992, 1997; Glickstein 2000). The lateral zone projects out (via dentate) to the red nucleus and to many cortical regions (via thalamus) including primary motor, premotor, parietal, and prefrontal cortices (Sasaki and others 1972; Asanuma and others 1983c; Orioli and Strick 1989; Kakei and others 1995; Clower and others 2001; Middleton and Strick 2001; Dum and Strick 2003). Thus, the lateral cerebellum may influence walking via cortical interactions and may be most important for voluntary modifications to the locomotor cycle.

Last, the flocculonodular lobe can be considered a separate functional region. This lobe receives inputs from the primary vestibular afferents, vestibular nuclei, and reticular nuclei (Clendenin and others 1974a, 1974b; Kotchabhakdi and Walberg 1978); the paraflocculus receives similar vestibular input and also visual inputs via pontine nuclei (Glickstein and others 1994). Purkinje cells in the flocculonodular lobe project out to directly influence the vestibular nuclei for the control of eye movements and balance (Ito and others 1976; Akaike 1983).

Cerebellar Functional Localization for the Control of Balance and Locomotion

Medial Zone and Flocculonodular Lobe

Influences of the flocculonodular lobe and the medial zone of the cerebellum on balance and locomotion are well documented in animals. Lesions of the vestibular or fastigial nuclei in cats and monkeys cause abnormal changes in upright postural tone (Sprague and Chambers 1953; Chambers and Sprague 1955a, 1955b), impairments in maintaining sitting and standing balance (Thach and others 1992), and difficulty walking (Sprague and Chambers 1953; Chambers and Sprague 1955a, 1955b; Thach and others 1992). Specifically, cats with damage to either the fastigial or vestibular nuclei produce a standing and walking posture of ipsilateral limb flexion and adduction combined with contralateral limb extension and abduction (Sprague and Chambers 1953; Chambers and Sprague 1955a, 1955b). Lesioned cats and monkeys have balance deficits that include impaired righting responses (Chambers and Sprague 1955a, 1955b) and frequent falls backward and toward the side of the lesion (Sprague and Chambers 1953; Chambers and Sprague 1955a, 1955b; Thach and others 1992). During locomotion, these animals also show abnormal timing of relative limb movements, reduced angular excursions of multiple leg joints, and decreased stride lengths (Yu and Eidelberg 1983).

Extracellular neural recordings are in agreement with the lesion data. In a series of studies, Orlovsky recorded from cells in the vestibular (Orlovsky 1972a, 1972b) and reticular (1972a) nuclei in decerebrate cats at rest and

during brainstem-microstimulated treadmill locomotion. Both cell types had increased tonic firing rates during locomotion compared to rest, and both showed frequency modulation timed to particular phases of the step cycle. Vestibular neurons were usually more active during the stance phase, with increased firing rates associated with increased extensor muscle activity. Reticular neurons were usually more active during the swing phase, with increased firing rates associated with increased flexor muscle activity (Orlovsky 1972a). Cerebellar influence over the vestibular nuclei was demonstrated by recording from these neurons before and after removal of the entire cerebellum (Orlovsky 1972b). Figure 1 shows the results of the study: There was an overall increase in firing rates in the vestibular nuclei and widespread loss of their modulatory characteristics after cerebellectomy (Orlovsky 1972b). Substantiating these results, Udo and colleagues (1976) found increased firing rates in the vestibular nuclei after cooling the vermis of cats, presumably due to loss of Purkinje cell inhibitory influences on the fastigius. In addition, they reported enhanced extensor muscle activity, increased ipsilateral stance phase durations, and somewhat abnormal joint kinematics. A later study showed that Purkinje cells in the cat cerebellar vermis also demonstrate frequency modulation related to specific phases of the step cycle, with many cells firing maximally during the late swing and early stance phases (Udo and others 1981). The neural activity has been speculated to relate to extensor muscle activity in preparation for foot touchdown.

Taken together, the physiological and lesion data indicate that the midline regions of the cerebellum (vermis and fastigial nuclei) and the flocculonodular lobe play a strong role in the control of balance and locomotion. A general role for these regions appears to be one of controlling extensor tone to maintain upright balance and stance and of exerting modulatory control over the rhythmic flexor and extensor muscle activations of the vestibular and reticular nuclei that generate portions of the locomotor pattern. In addition, the medial cerebellar region may also participate in altering the locomotor pattern based on sensory feedback from the limbs. Evidence for this comes from Udo and others (1982) who showed that vestibulospinal neural activity significantly increases when the dorsum of the stance phase contralateral foot is tapped. The increased activity is associated with a shortening of the perturbed stance phase and an earlier onset of the unperturbed (ipsilateral) limb extensor muscle activity, presumably to bring about earlier placement of the ipsilateral foot.

Intermediate Zone

The role of the intermediate region of the cerebellum in locomotion is not as well understood. Several studies have reported that walking deficits are less pronounced after damage to intermediate cerebellar regions compared to more medial regions (Chambers and Sprague 1955a, 1955b; Thach and others 1992). Upright posture

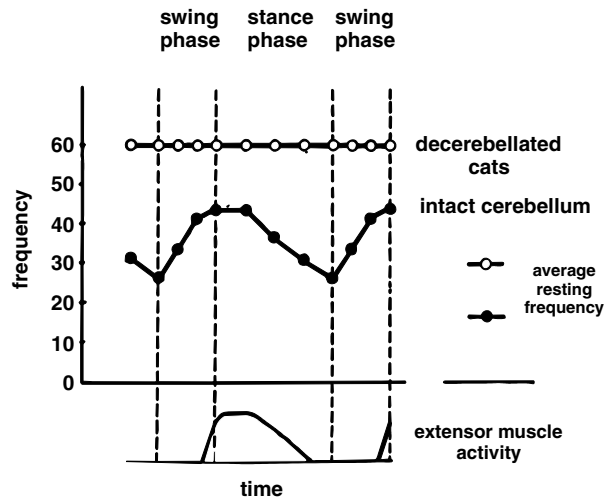

Fig. 1. Vestibular neurons lose their modulatory characteristics after cerebellectomy. Top two traces show discharge rates from a typical vestibular neuron during locomotion in decerebellate (open circles) and intact (filled circles) cats. Bottom trace shows a schematic of extensor muscle activity from the ipsilateral hindlimb during locomotion. Average discharge frequencies during rest are shown at right. From Orlovsky (1972b, p 89). Reproduced with permission from Elsevier.

and balance show little or no impairment during standing and walking in cats and monkeys with intermediate zone cerebellar damage (Chambers and Sprague 1955a, 1955b; Thach and others 1992). Cats with intermediate anterior lobe or interpositus lesions can often walk rather effectively over ground but have difficulty walking on a treadmill (Yu and Eidelberg 1983) or on rungs of a ladder (Chambers and Sprague 1955b). Some notable abnormalities of walking that have been reported include a loss of placing reflexes (Chambers and Sprague 1955a), ipsilateral limb hypermetria during swing phase (Chambers and Sprague 1955a, 1955b; Yu and Eidelberg 1983), and abnormal timing of forelimb stance and swing durations (Udo and others 1980). With respect to this latter impairment, lesions of the intermediate cerebellar cortex have been shown to result in an increased swing phase and a decreased stance phase in the ipsilateral limb and a decreased swing phase and an increased stance phase in the contralateral limb (Udo and others 1980). This pattern of movement is associated with enhanced ipsilateral flexor muscle activity in late stance and throughout swing phase (Udo and others 1979).

Recordings of simple spikes from Purkinje cells in the cat paravermal cortex (Armstrong and Edgley 1988; Edgley and Lidierth 1988) or from the interpositus nucleus (Armstrong and Edgley 1988) do show some step-cycle-related rhythmic modulation during locomotion. For most cells, the greatest activity from Purkinje cells occurs during the transition between ipsilateral stance and swing phases, and the least activity occurs during midstance (Edgley and Lidierth 1988). Recordings from the red nucleus (a major output target

of the interpositus and dentate nuclei) of thalamic cats show that these cells increase their firing rates during locomotion (Orlovsky 1972c). Their neural activity is timed to the phases of the step cycle, with most cells showing maximal firing frequencies during peak swing phase flexor muscle activity. If the entire cerebellum is subsequently removed, spontaneous discharge rates decrease overall but still show a relative increase during locomotion compared to resting. Probably more important, however, the previously noted frequency modulation during locomotion diminishes radically (Orlovsky 1972c). It is unclear, however, whether the cells in the intermediate cerebellar region directly regulate the rate and magnitude of muscle activity. Armstrong and Edgley (1988) found that discharge rates from the interpositus nuclei and Purkinje cells in the intermediate region increase only minimally when locomotor patterns changed from a slow to a fast speed or from level to inclined walking. In contrast, Schwartz and others (1987) showed that cells in the interpositus nuclei of decerebrate cats were highly modulated during treadmill locomotion and had firing patterns that correlated with either biceps or triceps muscle activity.

It is clear that the intermediate cerebellar region is involved in locomotor control, although in a different way than the medial cerebellar region. The consistent finding that over-ground locomotion is less impaired than treadmill or ladder walking (Yu and Eidelberg 1983; Chambers and Sprague 1955b) suggests that one main function of the intermediate region might relate to the specific control of limb placement (e.g., regulating the timing, amplitude, and trajectory of limb elevation and descent), particularly when more than the usual amount of precision is required. This idea is consistent with the findings of swing phase hypermetria (Chambers and Sprague 1955a, 1955b; Yu and Eidelberg 1983) and abnormal timing of stance and swing durations (Udo and others 1980) but a lack of any overt changes in balance or postural tone (Chambers and Sprague 1955a, 1955b; Thach and others 1992) after damage to the intermediate cerebellar region.

Lateral Zone

Typically, animal lesion studies show that over-ground walking deficits are minimal after lateral cerebellar damage (Chambers and Sprague 1955b; Thach and others 1992). One study reported that some lesioned monkeys produce large, bounding "grasshopper" leaps when attempting to walk (Botterell and Fulton 1938). However, these animals had large, acute lesions that often involved both intermediate and lateral zones of the cerebellar hemisphere. The most consistent finding from lesion studies is that animals can walk fairly normally over ground (Chambers and Sprague 1955b; Thach and others 1992), although with obvious misplacement of the ipsilateral limbs during other locomotor tasks, such as beam walking (Chambers and Sprague 1955b).

Electrophysiological studies indicate that the lateral cerebellar region is involved in the control of locomotion in certain circumstances. Cells in the dentate nucleus of decerebrate cats have little step-cycle-related modulation during normal treadmill locomotion but significant modulation during periods of a locomotor perturbation, for example, when the treadmill is suddenly stopped (Schwartz and others 1987). Purkinje cells in the lateral cortex and cells in the dentate nuclei of cats also show step-cycle-related rhythmical modulation during locomotion on horizontal rungs of a circular ladder (Marple-Horvat and Criado 1999). Some of these neurons also show frequency modulation according to unexpected movements of ladder rungs that are located just ahead of the walking cats (Marple-Horvat and others 1998). An example of such a dentate cell is shown in Figure 2. When a ladder rung located in front of the walking cat is elevated, the dentate neuron increases its firing rate. The increased firing is thought to reflect a response to a behaviorally relevant visual cue rather than a simple response to a visual stimulus. Evidence for this is that the magnitude of neural responses is scaled to the immediacy of the required motor response and that neural responses become markedly shorter and of much smaller amplitude when the cat is presented with visual cues that are irrelevant to the ongoing motor behavior (Marple-Horvat and others 1998).

These results indicate that the lateral cerebellar zone also contributes to the control of locomotion. The role, however, appears to be somewhat restricted to particular circumstances. The lateral cerebellum is especially active under novel walking conditions (Schwartz and others 1987), when precise limb placement is necessary, or when strong reliance on visual information is required (Marple-Horvat and others 1998; Marple-Horvat and Criado 1999). The motor cortex also produces rhythmic neural activity during locomotion in intact animals (Armstrong and Drew 1984), and its activity is greater when visual guidance is required (Drew 1988; Beloozerova and Sirota 1993). The lateral cerebellum may play an integral role in modulating motor cortical activity during visually guided locomotion because this region of the cerebellum receives abundant afferent projections from higher visual cortical areas and provides a major source of visual information to the motor cortex (Robinson and others 1984; Stein and Glickstein 1992).

In summary, medial cerebellar regions play a primary role in regulating extensor tone, sustaining upright stance and dynamic balance control, and modulating the rhythmic flexor and extensor muscle activity that makes up the locomotor pattern. In contrast, the intermediate cerebellar regions play a fairly minimal role in controlling stance and posture. Instead, these regions appear more important for directing limb placement and regulating agonist-antagonist muscle pairs to control the relative timing, amplitude, and trajectory of limb movements, especially in circumstances in which more precision is required. The lateral cerebellum is less important for the control of balance and uninterrupted level walking. However, lateral cerebellar regions do seem to play a significant role in making adjustments to the normal

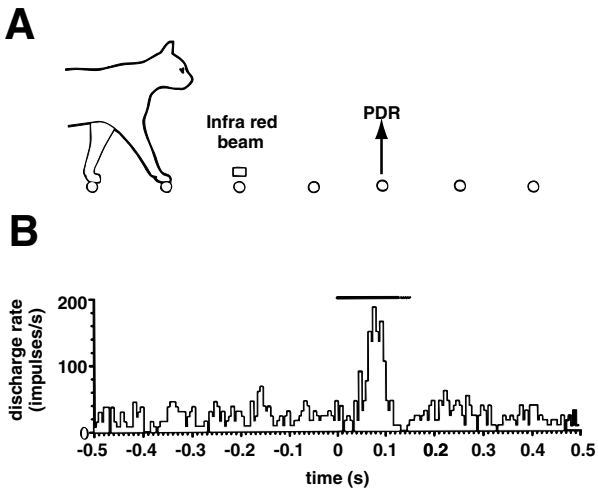

Fig. 2. Some dentate neurons show frequency modulation in response to a behaviorally relevant stimulus (e.g., the response to rung movement is greater than that seen in response to a visual stimulus alone or to rung movement that is irrelevant to the ongoing motor behavior). *A*, The basic paradigm. A predisplaceable rung (PDR) was made to move upward when triggered by cats' limbs crossing in front of an infrared beam aimed at the walkway. *B*, Discharge firing (averaged over 27 trials) from a dentate neuron in response to rung movement during locomotion (bin width, 5 ms). Horizontal bar indicates the duration of rung movement (starting at time 0). From Marple-Horvat and others (1998, p 491, 502). Reproduced with permission from the Physiological Society.

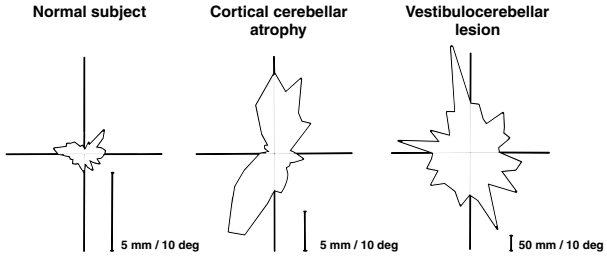

Fig. 3. Sway position (center of pressure) histograms (bin width 10 degrees polar; sampling frequency 30 ms; time 60 sec) from a typical control subject (*left*), a cerebellar subject with anterior lobe cortical atrophy (*middle*), and a cerebellar subject with a vestibulocerebellar medulloblastoma (*right*). Note the scale changes across plots. The cerebellar subject with anterior lobe atrophy shows moderate instability in the anterior-posterior dimension. The cerebellar subject with vestibulocerebellar damage shows severe instability that is omnidirectional. From Dichgans and Mauritz (1983, p 638). Reproduced with permission from Blackwell.

locomotor pattern in novel contexts or when strong visual guidance is required.

Cerebellar Control of Balance and Locomotion in Humans

Human studies of cerebellar involvement in balance and locomotion are more limited than the extensive animal literature. Cerebellar damage and disease occurs somewhat infrequently and is often accompanied by injury to other CNS regions such as the brainstem or spinal cord. This makes determination of the specific role of the cerebellum in balance and locomotor control more difficult. In addition, even focal cerebellar lesions in humans often span multiple longitudinal zones (Amarenco 1991), making it hard to localize functions to particular cerebellar regions. Nevertheless, studies of humans with cerebellar damage have provided important contributions to our understanding of cerebellar control of balance and locomotion.

Cerebellar damage in humans typically results in increased standing postural sway (Mauritz and others 1979; Diener and others 1984; Dichgans and Diener 1985). Mauritz, Dichgans, and colleagues have shown that cerebellar deficits of sway (center of pressure movements) are specific, depending on lesion location (Mauritz and others 1979; Dichgans and Mauritz 1983). Figure 3 shows center-of-pressure position histograms from a healthy subject, a subject with late anterior cerebellar cortical atrophy, and a subject with a vestibulocerebellar medulloblastoma (Dichgans and Mauritz 1983). The subject with anterior lobe atrophy has increased center-of-pressure movement in the anterior-posterior direction only, whereas the subject with the vestibulocerebellar lesion has increased movement that is omnidirectional. Overall, the authors report that balance deficits of subjects with anterior lobe damage are characterized by increased sway (of high velocity and low amplitude) in the anterior-posterior dimension, postural tremor, and increased intersegmental movements of the head, trunk, and legs. Localized vestibulocerebellar lesions lead to balance deficits characterized by increased postural sway (of low frequency and high amplitude) without a preferred direction and without increased intersegmental movements. Subjects with neocerebellar (lateral) lesions tend to have only slight postural instability that is indistinguishable from control subjects (Mauritz and others 1979; Dichgans and Mauritz 1983; Diener and others 1984).

Cerebellar damage in humans is also associated with hypermetric postural responses to surface displacements (Horak and Diener 1994) and an impaired ability to learn responses to predictable perturbations during quiet standing (Horak and Diener 1994) or step initiation (Timmann and Horak 1998). Figure 4*A* shows electromyographic activity, postural sway, shear forces, and surface torque produced by healthy subjects and individuals with anterior lobe cerebellar cortical atrophy when responding to forward body sway induced by a backward surface displacement (Horak and Diener 1994). Subjects with cerebellar damage show hypermetric postural responses and overshooting of the initial posture, with larger than normal surface-reactive torque responses and exaggerated and prolonged muscle activity. Abnormalities in learning this postural response will be discussed in the next section.

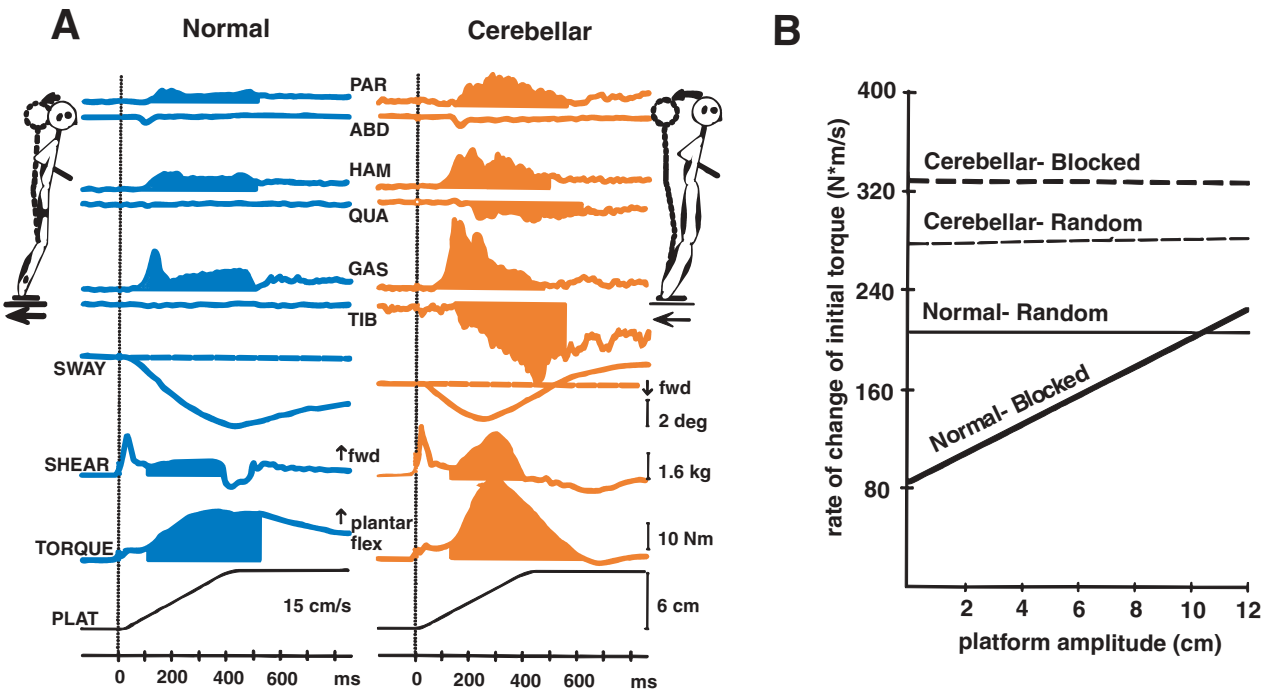

Fig. 4. *A*, Postural responses from control and cerebellar groups (average of 10 trials × 10 subjects in each group) after backward platform translations of 15 cm/sec for 6 cm. Traces show (top to bottom): electromyographic (EMG) recordings from paraspinals (PAR), rectus abdominus (ABD), biceps femoris (HAM), rectus femoris (QUA), gastrocnemius (GAS), and tibialis anterior (TIB) muscles; sway; shear forces; surface torque; and platform displacement. Shading indicates the first 400 ms of EMG activation and active surface reactive forces. Postural responses of cerebellar subjects are increased, with excessive and prolonged muscle activity, larger sway, and greater torque production. *B*, Average linear regression through all subjects' initial torque rates in response to randomized versus blocked (predictable) presentations of platform displacement amplitudes. Unlike healthy controls, subjects with cerebellar damage are not able to learn to scale their initial responses to predictable amplitudes of postural displacement. From Horak and Diener (1994, p 483, 490). Reproduced with permission from the American Physiological Society.

Physiological studies from animals suggest that cerebellar control of posture, equilibrium, and locomotion are tightly coupled and localized to the medial zone; control of complex, visually guided limb movements are relatively uncoupled to locomotion and are localized to the lateral zone (Botterell and Fulton 1938; Chambers and Sprague 1955a, 1955b; Arshavsky and others 1983; Yu and Eidelberg 1983). Human locomotor control, however, may be different. Because human locomotion is bipedal, it is inherently less stable (i.e., there are fewer contact points with the ground and a smaller base of support) than quadrupedal locomotion. Human gait therefore probably requires additional descending cerebral cortical control. Hence, bipedal locomotion may require additional contributions from the lateral cerebellum (which has heavy projections to and from cerebral cortical regions) that are less critical in quadrupedal locomotion. If this were true, we would expect that both balance and leg coordination deficits would contribute to the features of cerebellar gait ataxia.

We have examined the relative extents to which balance and voluntary leg coordination deficits contribute to cerebellar gait ataxia (Morton and Bastian 2003b). Figure 5*B* shows joint angles during walking for a healthy subject, a cerebellar subject with primarily balance deficits, and a cerebellar subject with primarily voluntary leg coordination deficits (see Fig. 5*A*). The cerebellar subject with severe balance deficits has several characteristic features of cerebellar gait ataxia, including reduced joint excursions, increased stride-to-stride variability in the joint angles, decomposition between the ankle and knee joints, and reduced walking speed. The cerebellar subject with severe leg coordination deficits does not show these signs of gait ataxia and, in fact, has leg kinematics that are indistinguishable from the control subject. We found that as a group, cerebellar subjects with only balance deficits demonstrate nearly all the classic features of gait ataxia (e.g., reduced stride lengths, increased stride widths, reduced joint excursions, abnormal swing foot trajectories, increased variability in foot placement, and joint-joint decomposition), but cerebellar subjects with only leg coordination deficits walk almost normally (Morton and Bastian 2003b). Therefore, during uninterrupted level walking, balance deficits contribute much more strongly to cerebellar gait ataxia than do visually guided leg control deficits.

Decomposition of movement is commonly seen in subjects with cerebellar gait ataxia (Hallett and Massaquoi 1993; Palliyath and others 1998; Earhart and Bastian 2001). Decomposition refers to the strategy of

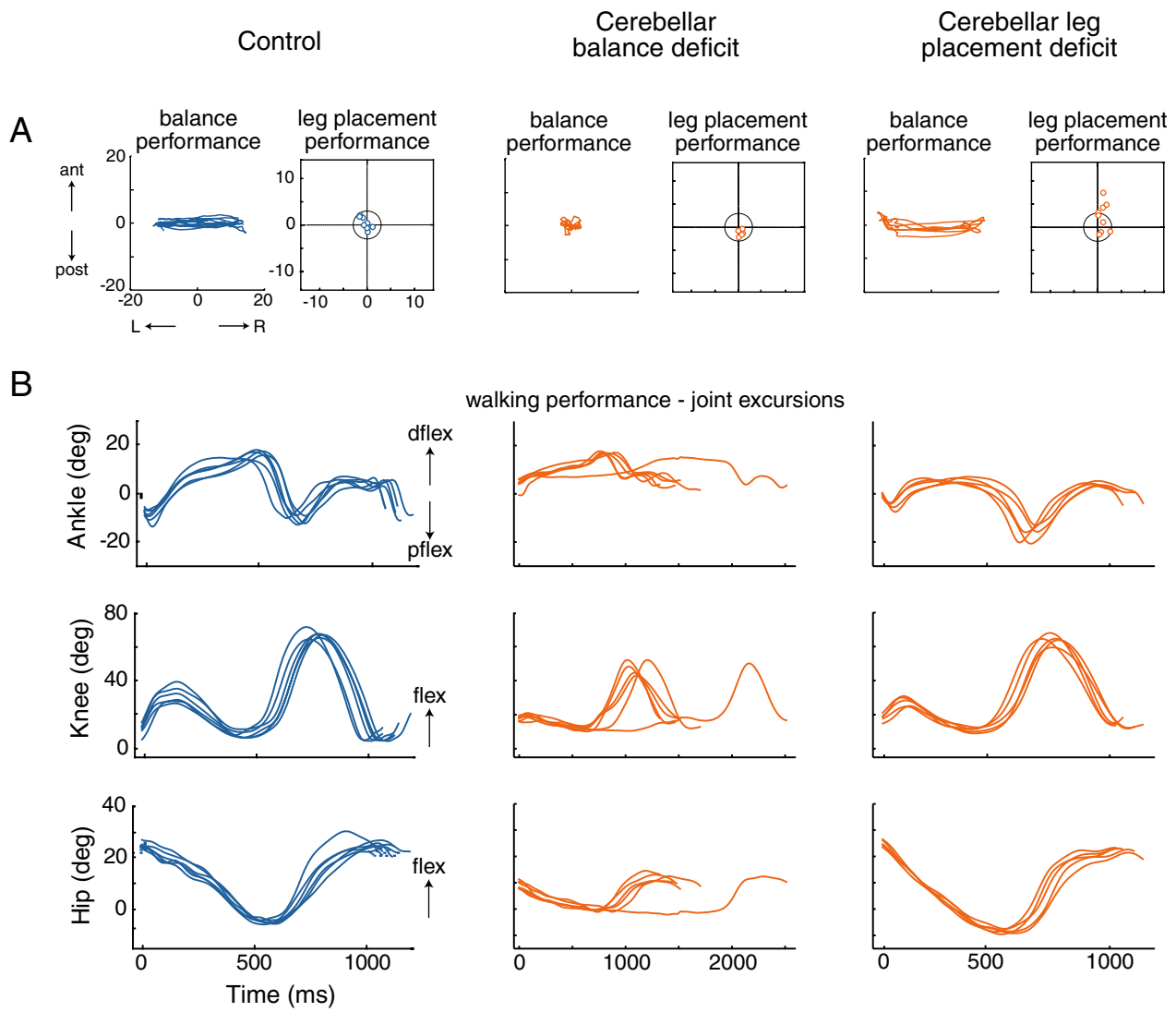

Fig. 5. *A*, Overhead view of performance on a balance task (center-of-pressure paths during lateral weight shifting) and a voluntary leg placement task (foot endpoint locations during targeted stepping) by a typical control subject, a cerebellar subject who has balance but not leg placement deficits, and a cerebellar subject who has leg placement but not balance deficits. *B*, Angular excursions at the ankle, knee, and hip joints during level walking for the same three subjects. Trials are plotted for single strides (several strides are overlaid for each subject) from heelstrike to heelstrike. Note the time scale changes across plots. The balance-impaired cerebellar subject shows many features of gait ataxia including reduced angular excursions, joint-joint decomposition, increased stride-to-stride variability, and reduced speed. The leg placement-impaired cerebellar subject (who has no significant balance deficit) walks similarly to the control subject. From Morton and Bastian (2003b, p 1850). Reproduced with permission from the American Physiological Society.

breaking down a multijoint movement, such as combined hip, knee, and ankle rotations, into a series of single-joint movements (Holmes 1917, 1939), presumably to compensate for impaired multijoint coordination. Figure 6*A* shows angle-angle plots for a healthy subject and an individual with a cerebellar stroke during the stance phase of walking on an inclined surface (Earhart and Bastian 2001). The cerebellar subject shows clear decomposition between the ankle and knee joints. This can be identified by the large horizontal and vertical components in the angle-angle plot, indicating long periods of movement at one joint while the other is kept fixed. Figure 6*B* depicts group data comparing the extent of ankle-knee decomposition in controls versus cerebellar subjects over level and various inclined surfaces. Overall, subjects with cerebellar damage have increased joint-joint decomposition that tends to worsen as the surface inclination increases (Earhart and Bastian 2001).

Decomposition of leg movements during gait may be an attempt to compensate for poor control of intersegmental dynamics, although this issue has not yet been addressed in the literature. Several studies have shown that cerebellar damage causes specific deficits in the control of interaction torques (those torques produced at a segment that are caused by movements at other linked segments) during reaching movements (Bastian and oth-

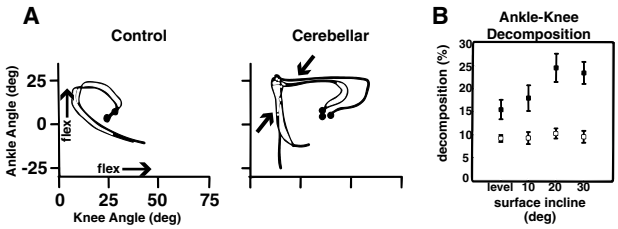

Fig. 6. *A*, Angle-angle plots depict ankle angle relative to knee angle movements from several trials from a typical control subject (*left*) and a cerebellar subject with an infarct of the superior cerebellar artery (*right*) during stance phase of walking over a 30-degree incline. Circles show stance phase onset. Arrows on the cerebellar plot show the decomposition (e.g., when one joint moves while the other is kept fixed), indicated by the horizontal and vertical components of the trace. *B*, Average (±1 *SE*) ankle-knee decomposition for control (open squares) versus cerebellar (filled squares) groups during stance phase walking over level and inclined surfaces. Decomposition is quantified as the percentage of time in stance that the ankle or knee joint is kept still (angular velocity < 5 degrees/sec) while the other joint is moving (angular velocity ≥ 5 degrees/sec). Cerebellar subjects show increased decomposition overall compared to controls, with worsening effects as the surface incline rises. From Earhart and Bastian (2001, p 764). Reproduced with permission from the American Physiological Society.

ers 1996; Topka and others 1998b; Bastian and others 2000; Cooper and others 2000). Hypermetria, or overshooting of targets, and curved hand paths are hallmarks of cerebellar dysfunction (Holmes 1939; Hallett and Massaquoi 1993; Bastian and others 1996; Topka and others 1998a; Bastian and others 2000), which have been shown to be related to impaired control of interaction torques during reaching (Bastian and others 1996; Topka and others 1998b; Bastian and others 2000; Cooper and others 2000). We have examined the relationship between cerebellar leg hypermetria and control of intersegmental dynamics during obstacle avoidance. We found that cerebellar leg hypermetria (excessive foot elevation) during stepping in place over a small obstacle is related to an overcompensation for interaction torques at the knee (Morton and others 2004). No group has yet examined whether certain features of cerebellar gait ataxia (e.g., abnormal and irregular foot placement, excessive foot path curvature, impaired interjoint coordination of the stance limb) may also be related to poor control of intersegmental dynamics.

Cerebellar Contributions to Motor Learning

It is well known that cerebellar damage affects motor adaptation (error-based learning) of several types of specific arm (Deuschl and others 1996; Martin and others 1996; Lang and Bastian 1999) and eye movements (Robinson 1976; Raymond and others 1996; Krupa and Thompson 1997; Takagi and others 1998), although relatively few studies have addressed adaptation during locomotion. The spinal cord alone is capable of producing some relatively simple, short-term changes in locomotor output based on sensory feedback (Forssberg and others 1975), but longer term and more complicated adaptations of the locomotor pattern probably depend on supraspinal structures such as the cerebellum (Grillner and Wallen 1985; Armstrong 1986). In the cat, the ability to adapt to perturbations during treadmill locomotion is impaired following nitric oxide deprivation (which abolishes long-term depression in the cerebellar cortex) in lobules V and VI of the vermis (Yanagihara and Kondo 1996). Only a few studies have addressed adaptation of posture and locomotion in humans.

Horak and Diener (1994) studied whether cerebellar subjects could learn to adjust for predictable postural perturbations during standing. Figure 4*B* shows the rate of change in initial torque responses for control and cerebellar groups during randomized versus blocked presentations of backward surface perturbations of different displacement amplitudes. When perturbation amplitudes are presented in a blocked (predictable) format, healthy subjects are able to learn to appropriately scale their initial postural responses. In contrast, cerebellar subjects are unable to learn to use predictive feedforward control to scale their postural responses to expected perturbation amplitudes (Horak and Diener 1994).

A few studies suggest that cerebellar damage impairs certain types of locomotor adaptation. Rand and colleagues (1998) studied adaptation to sudden but predictable changes in treadmill speed during walking in subjects with cerebellar damage and mild ataxia compared to controls. Results suggested that cerebellar subjects can adapt to changes in speed, but they have considerable variability in the timing and duration of muscle activation patterns and use a strategy that is different than that used by healthy subjects. Another interesting paradigm capitalized on the fact that precise control of the rotations of the trunk relative to the stance and swing limbs is required to maintain a straight walking trajectory. After stepping in place on a rotating disc with the body stationary in space, healthy blindfolded subjects walk in a curved rather than a straight trajectory, an aftereffect referred to as "podokinetic after-rotation" (Gordon and others 1995). This occurs because stepping on the rotating disc produces a recalibration of the proprioceptive gain between the trunk and stance limb rotations. Subjects with cerebellar damage, however, show reduced amplitudes of the after-rotation (Earhart and others 2002), indicating a diminished capacity to store a novel body orientation during locomotion. In a third

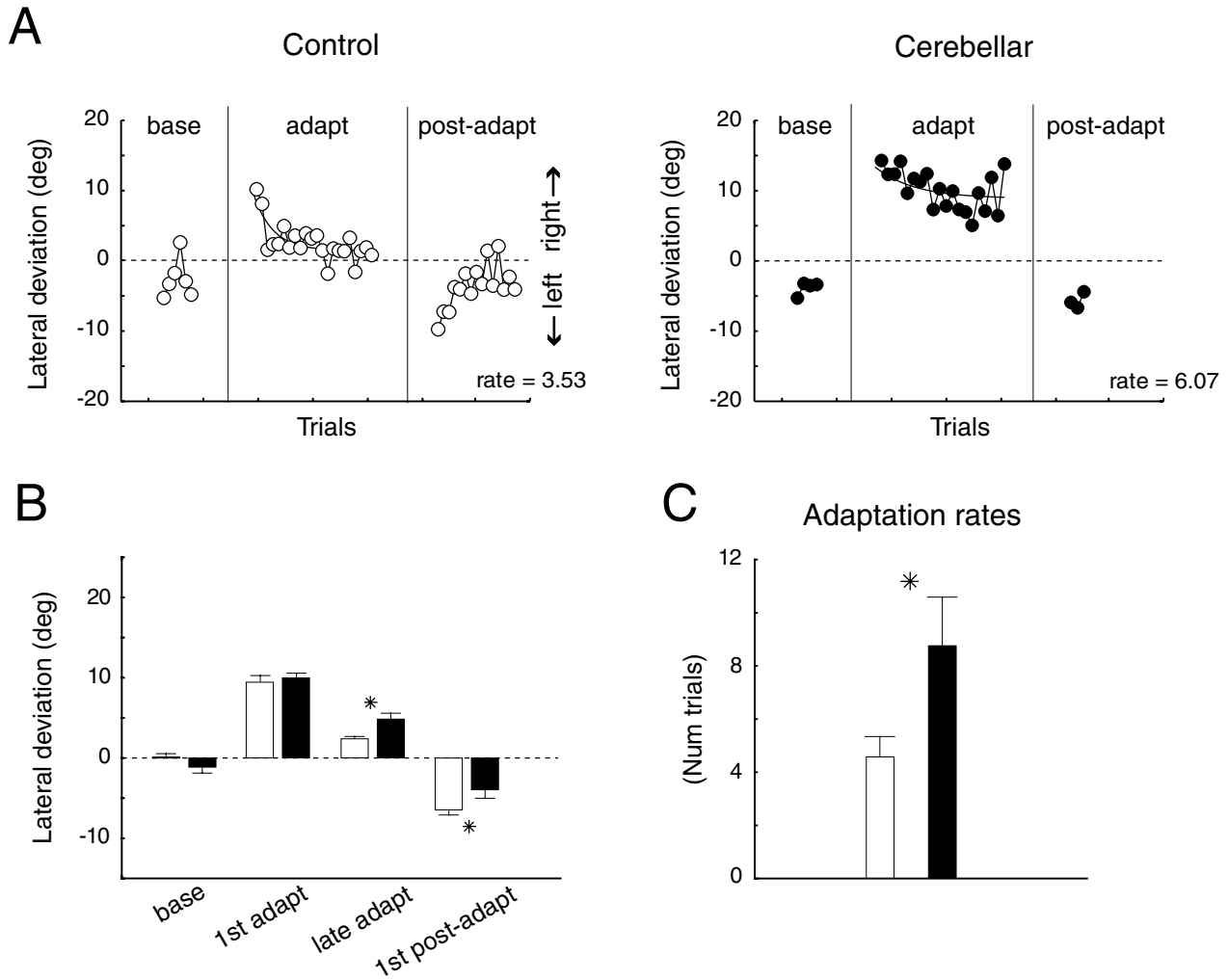

Fig. 7. *A*, Maximal lateral deviations of the walking trajectory for a control subject (open circles) and a cerebellar subject with a stroke involving the superior cerebellar artery (filled circles) plotted for each walking trial before (baseline), during (adaptation), and after (postadaptation) prism exposure while walking. *B*, Average (±1 *SE*) lateral walking deviations for control (open bars) versus cerebellar (filled bars) groups. Bars show deviations averaged over all baseline trials, the first adaptation trial, the average of the last five adaptation trials, and the first postadaptation trial. Cerebellar subjects show a reduced adaptation magnitude and a diminished negative aftereffect after prism walking adaptation. *C*, Average (±1 *SE*) adaptation rates (the time constant from exponential functions fit to lateral walking deviations during prism adaptation) for control versus cerebellar groups. Cerebellar subjects show increased rates of prism walking adaptation.

locomotor adaptation study, we demonstrate that subjects with cerebellar damage are impaired in adapting the locomotor trajectory to novel visual input (Morton and Bastian 2003a). After only a few trials of walking while wearing rightward laterally displacing prism glasses, healthy subjects adapt their walking path to maintain a straight trajectory despite the conflicting visual information from the prisms. Figure 7*A* shows trial-by-trial changes in the walking direction during adaptation to prisms for a healthy control and a subject with cerebellar damage. The cerebellar subject does not improve his walking direction to the same extent or at the same rate as the control subject. As a group (Fig. 7*B*), subjects with cerebellar damage show a reduced extent of adaptation, increased rate of adaptation (Fig. 7*C*), and smaller capacity for storage of the adaptation (Morton and Bastian 2003a). In sum, the cerebellum seems to be important for adapting posture and locomotion through trial-and-error practice, though admittedly, the data for this are somewhat limited.

Future Research

Despite many years of research, there is no consensus on the basic operation of the cerebellum. Given the very regular anatomical structure of the cerebellar cortex, it seems likely that the different zones of the cerebellum are performing the same basic computation, although on different inputs. Some theories of cerebellar function include timing control (Braitenberg 1967; Llinas and Sasaki 1989; Ivry and others 1988), gain control (Horak and Diener 1994; Timmann and Horak 1998), multijoint

coordination and combination (Thach and others 1992), dynamics control (Bastian and others 1996, 2000), and internal models of limb dynamics (Schweighofer and others 1998; Wolpert and Kawato 1998). Many of these ideas are based on studies of only one or a few types of movement. To be satisfying, a theory of cerebellar function would have to explain the wide variety of movement deficits associated with cerebellar damage (including balance and gait abnormalities, eye movement deficits, limb ataxia, and speech impairments) in addition to its roles in learning movement and various cognitive processes. Uncovering the primary computation that the cerebellum performs is one of the greatest challenges facing cerebellar researchers today.

There are also practical questions to be answered about cerebellar control of balance and locomotion. For example, we have found that cerebellar gait ataxia is more strongly related to balance deficits than voluntary leg control deficits (Morton and Bastian 2003b). This finding suggests that rehabilitation for cerebellar gait ataxia might be most beneficial if directed at balance training rather than specific exercises for the coordination of leg movements. Few studies of rehabilitation training for cerebellar ataxia exist, but one case study suggests that balance training can have a modest but positive effect on walking (Gill-Body and others 1997). Larger outcome studies should be conducted to determine whether this type of training might be effective for people with cerebellar gait ataxia.

Of course, an important issue to consider regarding rehabilitation training is whether cerebellar subjects will be able to benefit from practice. Abundant evidence indicates that the cerebellum is critical for practice-dependent motor learning (Weiner and others 1983; Deuschl and others 1996; Martin and others 1996; Yanigahara and Kondo 1996; Baizer and others 1999; Lang and Bastian 1999; Earhart and others 2002). This suggests that patients with cerebellar damage would be unlikely to benefit from rehabilitation and training. However, it is not known whether, if given a greater duration or intensity of practice, cerebellar patients could learn new movements. It may be that the effectiveness of training would depend on the location and extent of cerebellar damage. Understanding this issue is of great importance because rehabilitation training is one of the only treatments currently available for patients with cerebellar movement disorders.

A related question is whether there are certain specific types of movements that patients with cerebellar damage can successfully learn. For instance, if the cerebellum is disproportionately involved in control of complex multijoint movements, then perhaps cerebellar subjects might be capable of adapting simpler movements that involve control of fewer segments at a time (Bastian and others 2000). Alternatively, it may be that some changes in motor behavior that do not require trial-and-error learning may be effective means of training patients with cerebellar damage. For example, training new behaviors that use habit formation (Jog and others 1999; Witt and others 2002) or reward-based mechanisms (Schultz and others 1997) may rely more heavily on other brain regions such as the basal ganglia and would thus be largely unaffected by cerebellar damage.

Summary

Grillner and Wallen (1985) have proposed that there are three primary requirements of the CNS to produce normal patterns of locomotion. First, the CNS must generate the basic pattern of rhythmic reciprocal flexor and extensor muscle activity that makes up the basic locomotor synergy. Second, the CNS must maintain control of equilibrium. The animal must be able to keep its center of mass over a constantly moving base of support to maintain the body upright in space. Finally, the CNS must have some adaptive capabilities for locomotor control so that the basic pattern can be altered according to novel contexts or changes in the behavioral goal.

We propose that the cerebellum contributes to each of these requirements in specific ways. First, although the basic locomotor synergy is likely to be generated in the brainstem and spinal cord (Grillner 1975; Forssberg and others 1980), the cerebellum probably contributes to modulations of the timing, rate, and force of muscle activity controlling locomotor patterns (Orlovsky 1972a, 1972b; Udo and others 1976, 1979; Schwartz and others 1987; Edgley and Lidierth 1988). Thus, the cerebellum helps coordinate relative motions at various joint segments within a limb (Yu and Eidelberg 1983; Hallett and Massaquoi 1993; Palliyath and others 1998; Earhart and Bastian 2001; Morton and others 2004) and probably between limbs as well (Udo and others 1980; Yanigahara and Kondo 1996; Bastian and others 1998) to produce the highly stereotyped walking pattern, with smooth foot trajectories, accurate foot placements, and coordinated motions among the hip, knee, and ankle joints and between the right and left legs.

Second, the cerebellum plays an integral role in the control of upright posture during walking. Cerebellar damage causes impaired sitting, standing, and walking balance; abnormal postural tone; and frequent falling during locomotion (Sprague and Chambers 1953; Chambers and Sprague 1955a, 1955b; Thach and others 1992; Mauritz and others 1979; Diener and others 1984; Dichgans and Diener 1985). The cerebellum uses both anticipatory (Timmann and Horak 1998) and feedback (Horak and Diener 1994) mechanisms to maintain balance during locomotion, and when those mechanisms fail, gait ataxia results (Morton and Bastian 2003b).

Finally, the cerebellum may be important for providing adaptability to the locomotor pattern. Cerebellar damage can interfere with both short-term adjustments based on external sensory feedback, such as tapping the dorsum of the paw (Udo and others 1982), and more complex motor adaptations based on trial-and-error practice (Yanigahara and Kondo 1996). Cerebellar patients are impaired in adapting to perturbations during treadmill walking (Rand and others 1998) or platform standing (Timmann and Horak 1998). They are also

compromised in learning novel sensorimotor (Earhart and others 2002) or visuomotor recalibrations (Morton and Bastian 2003a) of the locomotor trajectory.

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